

Euler–Lagrange equation generalisation for n arm pendulum

General equation format

$$M \cdot \ddot{\vec{\theta}} = \vec{h} \quad (1)$$

Matrix M

Derived equation for 2 arm pendulum

$$\begin{pmatrix} (m_1 + m_2)l_1^2 & m_2l_1l_2 \cos(\theta_1 - \theta_2) \\ m_2l_1l_2 \cos(\theta_1 - \theta_2) & m_2l_2^2 \end{pmatrix} \quad (2)$$

Derived equation for 3 arm pendulum

$$\begin{pmatrix} (m_1 + m_2 + m_3)l_1^2 & (m_2 + m_3)l_1l_2 \cos(\theta_1 - \theta_2) & m_3l_1l_3 \cos(\theta_1 - \theta_3) \\ (m_2 + m_3)l_1l_2 \cos(\theta_1 - \theta_2) & (m_2 + m_3)l_2^2 & m_3l_2l_3 \cos(\theta_2 - \theta_3) \\ m_3l_1l_3 \cos(\theta_1 - \theta_3) & m_3l_2l_3 \cos(\theta_2 - \theta_3) & m_3l_3^2 \end{pmatrix} \quad (3)$$

$$\begin{pmatrix} (m_1 + m_2 + m_3)l_1l_1 \cos(\theta_1 - \theta_1) & (m_2 + m_3)l_1l_2 \cos(\theta_1 - \theta_2) & m_3l_1l_3 \cos(\theta_1 - \theta_3) \\ (m_2 + m_3)l_1l_2 \cos(\theta_1 - \theta_2) & (m_2 + m_3)l_2l_2 \cos(\theta_2 - \theta_2) & m_3l_2l_3 \cos(\theta_2 - \theta_3) \\ m_3l_1l_3 \cos(\theta_1 - \theta_3) & m_3l_2l_3 \cos(\theta_2 - \theta_3) & m_3l_3l_3 \cos(\theta_3 - \theta_3) \end{pmatrix} \quad (4)$$

In General

$$M_{ij} = \left(\sum_{k=\max(i,j)}^n m_k \right) l_i l_j \cos(\theta_i - \theta_j)$$

Vector $\vec{h}$

Derived equation for 2 arm pendulum

$$\begin{pmatrix} -m_2l_1l_2\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - gl_1(m_1 + m_2) \sin \theta_1 \\ m_2l_1l_2\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - m_2gl_2 \sin \theta_2 \end{pmatrix} \quad (5)$$

Derived equation for 3 arm pendulum

$$\begin{pmatrix} -(m_2 + m_3)l_1l_2\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - m_3l_1l_3\dot{\theta}_3^2 \sin(\theta_1 - \theta_3) - gl_1(m_1 + m_2 + m_3) \sin \theta_1 \\ (m_2 + m_3)l_1l_2\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - m_3l_2l_3\dot{\theta}_3^2 \sin(\theta_2 - \theta_3) - gl_2(m_2 + m_3) \sin \theta_2 \\ m_3l_1l_3\dot{\theta}_1^2 \sin(\theta_1 - \theta_3) + m_3l_2l_3\dot{\theta}_2^2 \sin(\theta_2 - \theta_3) - gm_3l_3 \sin \theta_3 \end{pmatrix} \quad (6)$$

Reordering arguments in sin to make those terms positive and adding missing terms for sin(0)

$$\begin{pmatrix} (m_1 + m_2 + m_3)l_1l_1\dot{\theta}_1^2 \sin(\theta_1 - \theta_1) - (m_2 + m_3)l_1l_2\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - m_3l_1l_3\dot{\theta}_3^2 \sin(\theta_1 - \theta_3) \\ - gl_1(m_1 + m_2 + m_3) \sin \theta_1 \\ (m_2 + m_3)l_1l_2\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + (m_2 + m_3)l_2l_2\dot{\theta}_2^2 \sin(\theta_2 - \theta_2) - m_3l_2l_3\dot{\theta}_3^2 \sin(\theta_2 - \theta_3) \\ - gl_2(m_2 + m_3) \sin \theta_2 \\ m_3l_1l_3\dot{\theta}_1^2 \sin(\theta_1 - \theta_3) + m_3l_2l_3\dot{\theta}_2^2 \sin(\theta_2 - \theta_3) + m_3l_3l_3\dot{\theta}_3^2 \sin(\theta_3 - \theta_3) \\ - gl_3m_3 \sin \theta_3 \end{pmatrix} \quad (7)$$

In General

$$\vec{h}_i = \left(\sum_{k=1}^n \left(\sum_{l=\max(i,k)}^n m_l \right) l_i l_k \dot{\theta}_k^2 \sin(\theta_k - \theta_i) \right) - g \left(\sum_{k=i}^n m_k \right) l_i \sin(\theta_i) \quad (8)$$